

2.3 Rank-Nullity Theorem

2.3.1 Nullity and Rank

Definition 2.3.1 — Nullity

Let $f : V \rightarrow W$ be a linear transformation. The **nullity** of f is defined by

$$\text{nullity}(f) := \dim(\ker(f)).$$

Definition 2.3.2 — Rank

Let $f : V \rightarrow W$ be a linear transformation. The **rank** of f is defined by

$$\text{rank}(f) := \dim(\text{im}(f)).$$

Example.

Recall that for $D : \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$ defined by

$$D(f(x)) = f'(x),$$

we have $\ker(D) = \mathbb{R}$ and $\text{im}(D) = \mathbb{R}[x]_{\leq 2}$. Compute the nullity and image of D .

Solution. $\text{nullity}(D) := \dim_{\mathbb{R}}(\ker(D)) = \dim_{\mathbb{R}}(\mathbb{R}) = 1.$

$\text{rank}(D) := \dim_{\mathbb{R}}(\text{im}(D)) = \dim_{\mathbb{R}}(\mathbb{R}[x]_{\leq 2}) = 3.$

Remark. For other examples of nullity and rank, review the examples in the last section, which we have it covered.

2.3.2 The Theorem

Theorem 2.3.1 — Rank-Nullity Theorem

Let V be a finite dimensional vector space and $f : V \rightarrow W$ be a linear transformation. Then we have

$$\text{nullity}(f) + \text{rank}(f) = \dim(V).$$

Proof. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a basis of $\ker(f)$, having $\text{nullity}(f) = \dim(\ker f) = n$. Since $\ker(f) \subseteq V$, by Basis Extension Theorem, we can extend S to $S' = \{\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{s}_1, \dots, \mathbf{s}_k\}$, a basis of V , having $\dim(V) = n + k$.

Claim. $S'' = \{f(\mathbf{s}_1), f(\mathbf{s}_2), \dots, f(\mathbf{s}_k)\}$ is a basis of $\text{im}(f)$.

Proof. We verify two properties.

Spanning. Let $\mathbf{w} \in \text{im}(f)$, we have $\mathbf{w} = f(\mathbf{v})$ for some $\mathbf{v} \in V$ by definition. As S' is a basis of V , we can write $\mathbf{v} = \sum_{i=1}^n \alpha_i \mathbf{v}_i + \sum_{i=1}^k \beta_i \mathbf{s}_i$. Then we apply f to both sides:

$$\mathbf{w} = f(\mathbf{v}) = f\left(\sum \alpha_i \mathbf{v}_i + \sum \beta_i \mathbf{s}_i\right) = \sum \alpha_i f(\mathbf{v}_i) + \sum \beta_i f(\mathbf{s}_i) = \sum \beta_i f(\mathbf{s}_i),$$

where the last equality holds since $\mathbf{v}_i \in \ker(f)$. Since $\mathbf{w}$ can be expressed by $f(\mathbf{s}_i)$'s, therefore S'' spans $\text{im}(f)$.

Linear independence. Suppose that $\alpha_1 f(\mathbf{s}_1) + \cdots + \lambda_k f(\mathbf{s}_k) = \mathbf{0}$. By linearity of f , it becomes $f(\alpha_1 \mathbf{s}_1 + \cdots + \alpha_k \mathbf{s}_k) = \mathbf{0}$. Therefore we have $\alpha_1 \mathbf{s}_1 + \cdots + \alpha_k \mathbf{s}_k \in \ker(f)$. Moreover, as S is a basis of $\ker(f)$, we know that $\sum \alpha_i \mathbf{s}_i = \sum \beta_i \mathbf{v}_i$ for some $\beta_i \in F$. Reordering gives $\sum \alpha_i \mathbf{s}_i - \sum \beta_i \mathbf{v}_i = \mathbf{0}$. As S' is linearly independent, this forces all α_i 's and β_i 's to be 0. This completes the proof of the claim. ■

Hence, we have

$$\text{rank}(f) = \dim(\text{im}(f)) = k = (n + k) - n = \dim(V) - \text{nullity}(f).$$
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